RESOLUTION MSC.33(63)
adopted on 23 May 1994
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ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CONVENTION ON STANDARDS OF TRAINING. CERTIFICA TION AND WATCHKEEPING FOR SEAFARERS, 1978
THE MARITIME SAFETY COMMITTEE.
RECALLING Article 28(b) of the Conventio n on the Interna tional Maritime Organization concerning the fUn c tion s of the Committee.
RECALLING FURTHER article XII(l)(a) of the Inte rnation al Conven tion on Standards of Tra i ning. Certification and Watchkeepin g for Sea fa rers. 1976, hereinafter referred to as "the Convention", concern i ng the proced ures for amending t he Annex to the Convention.
HAVING CONSIDERED, at its sixty-third session. amendments to the Conventi on proposed and circulated in accordance with article XII(l)(a)(i) thereof,
1. ADOPTS. in ac cordance with article XII (l) (a)(i v) of the Convention, amendments to the Conv ention , the text of whi ch is s et out in the Annex to the present resolution;

thereof,
1. ADOPTS. in ac cordance with article XII (l) (a)(i v) of the Convention, amendments to the Conv ention , the text of whi ch is s et out in the Annex to the present resolution;
2. DETERMINES, in accordance with a rticle XII(1)(a)(vii)2 of the Convention, that the amendments shall be deemed to have been acce pted on 1 July 19 95 unles s, prior to that date, more than one third of Parti es to the Convention or Partie s . the combined merchant fleets of whi ch constitute not less than fifty per cent of the gross tonnage of the world' s merchant shipping of ships of 100 gross register tons or more , have notified their objections to the amendment s;
3. INVITES Parties to note that , i n accordance with a rticle XII(l)(a)(ix) of the Convention, the amendments shall enter into f or ce on I January 1996 upon their acceptance in accordance with paragraph 2 above:
4. REQUESTS the Secretary-Genera l, in conformity with article XII(l)(a)(v) of the Convention, to transmit certified copies of the present resolution and the tex t of the amendments contained in the Annex to all Pa r ties to the Convention for a cceptance :

of the Convention, to transmit certified copies of the present resolution and the tex t of the amendments contained in the Annex to all Pa r ties to the Convention for a cceptance :

5. FVRTHER REQUESTS the Secretary - General to transmit copie s of the resolution and its Annex to Members of the Organization which are not Parties to the Convention.

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ANNEX

AMENDMENTS TO THE INTERNATIONAL CONVENTION ON STANDARDS OF TRAINING, CERTIFICA~ION AND WATCHKEEPING FOR SEAFARERS, 1978

A The existing chapter V is replaced by the following:

"CHAPTER V

SPECIAL TRAINING REQUIREMENTS FOR PERSONNEL ON TANKERS

Regulation VI I

Mann~~orv miDiml~ requirements for the t r ain i ng and qualifications of masters. Qfficer ~ and ratings of tankers

1 Officers and ratings who are to have specific du t ies , and responsibilities related to those duties, in connectio n wi t h cargo and cargo equipment on tankers shall have completed an appropriate sho re- based fire-fighting course: and

1 Officers and ratings who are to have specific du t ies , and responsibilities related to those duties, in connectio n wi t h cargo and cargo equipment on tankers shall have completed an appropriate sho re- based fire-fighting course: and

.1 at least three months of supervised sea servi c e on tank er s in order to acquire adequate knowledge of safe operationa l practic es ; or

.2 an approved tanker familiarization course covering at l east the syllabus given at Appendix 1 to this regu l ation.

The Administration may, however, accept a period of supervise d sea service shorter than three months as prescribed in subparagrap h .1 for officers and ratings of a tanxer. if all of the following conditions are met:

. 3 the period so accepted is not less than one month;

.4 the tanker is less than 1.600 gross tonnage:

.5 the duration of a voyage on which the tanker is engaged does not exceed 72 hours; and

. 6 the Administration is satisfied that the operational characteristics of the ship, including the number of voyages and number of loading and discharging operations in whi ch the ship is engaged during such

exceed 72 hours; and
. 6 the Administration is satisfied that the operational characteristics of the ship, including the number of voyages and number of loading and discharging operations in whi ch the ship is engaged during such period. are such that the same lev e l of knowledge may be acquired within that period.
2 Masters. chief engineer officers . chi e f mates. second engineer officers and. if other than the foregoing, any person with the immediate responsibi l ity for loading. diSCharging and care in transit or handling of cargo. in addition to the provisions of paragraph 1.1 and 1.2, shall have:
.1 experience appropriate to their du ties on tankers and relevant to the type of tanker on which they serve. i.e. oil tanker. chemical tanker or liquefied gas tanker; and
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.2 completed an approved specialized training programme appropriate to their duties on the type of tanker on which they serve, i.e. oi l tanker. chemical tanker or liquefied gas tanker. The specialized

.2 completed an approved specialized training programme appropriate to their duties on the type of tanker on which they serve, i.e. oi l tanker. chemical tanker or liquefied gas tanker. The specialized training programme shall cover at least the syllabus given at Appendix 2 or 3 or 4 to this regulation. as appropriate.

3 Within two years after the entry into force of the Convention for a Party, a seafarer may be considered to have met the requirements of paragraph 2.2 if he has served in a relevant capacity on board the type of tanker concerned for a period of not less than one year within the preceding five years.

4 Administrations shall ensure that an authorized document is issued to Officers and ratings, who are qualified in accordance with this regulation paragraph I or 2 as appropriate. or that any appropriate existing document is duly endorsed.

Appendix 1 to regulation VII

Tanker fnmiliarizAtion course

The tanker familiarization course referred to in paragraph 1.2 of regulation VII sball cover at least the syllabus given below.

1 Characteristics of cargoes

is duly endorsed.

Appendix 1 to regulation VII

Tanker fnmiliarizAtion course

The tanker familiarization course referred to in paragraph 1.2 of regulation VII sball cover at least the syllabus given below.

1 Characteristics of cargoes

An outline treatment including practical demonstration of the physical properties of oil. chemicals or gases carried in bulk; vapour pressureltemperature relationship. Influence of pressure on boiling temperature. Explanation of saturated vapour pressure, diffusion, partial pressure. flammability limit. flashpoint and auto-ignition temperature. Practical significance of flashpoint and lower flftmmable limit. Simple explanation of types of electrostatic charge generation. Chemical symbols and structures, elements of the chemistry of acids and bases. chemical reactions of well-known groupings, sufficient to enable proper utilization of codes.

2 Toxicity

Simple principles and explanation of basic concepts; toxicity limits, both acute and chronic effects of toxicity. systemic poisons and irritants.

3 Hazards

.1 Explosion and flammability hazards

Flammability limits . Sources of ignition and explosion .

2 Health hazards

both acute and chronic effects of toxicity. systemic poisons and irritants.
3 Hazards
.1 Explosion and flammability hazards
Flammability limits . Sources of ignition and explosion .
2 Health hazards
Dangers of skin contact, inhalation and ingestion.
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.3 Hazards to the environment
Effect on humaD and marine life of release of oil,
gases. Effect of specific gravity and soiubility.
vapour cloud drift. Effect of vapour pressure and
conditions
chemical s or
Danger from
atmospheric
. 4 Reactivity hazards, self-reaction, polymerization. effe c ts of
temperature, impurities as catalyst s . Reaction with air. water
and other chemicals .
. 5 Corrosion hazard s
Dangers to personnel. attacks on constructional material s.
Effects of concentration.
4 Hazard control
Inerting. water padding. drying agents, monitoring teChnique s.
Anti - static measures. Ventilation. Segregation . Cargo inhibiti on.
The importance of compatibility of materials.

Effects of concentration.
4 Hazard control
Inerting. water padding. drying agents, monitoring teChnique s.
Anti - static measures. Ventilation. Segregation . Cargo inhibiti on.
The importance of compatibility of materials.
5 Safety equipment and protection of personnel
The function and calibration of measuring instruments and similar equipment. Specialized fire - extinguishing appliances, breathing and escape apparatus. Safe use of protective clothing and equipment.
6 Pollution prevention
Procedures to be follo~ed to prevent air and water pollution and measures to be taken in the event of spillage, including the need to repor t immediately all relevant information to the appropriate officials when a spill is detected or when a malfunction has occurred which poses a risk of a spill; to notify promptly shore-based response personnel; and to implement properly s hipboard spill mitigation procedures.
Appendix 2 to regulation VII
Oil tankers
The specialized training programme referred to in paragraph 2.2 of regulation VI} appropriate to duties on oil tankers sha l l provide theoretical and practical knowledge of the following subjects:

Appendix 2 to regulation VII
Oil tankers
The specialized training programme referred to in paragraph 2.2 of regulation VI} appropriate to duties on oil tankers sha l l provide theoretical and practical knowledge of the following subjects:
.1 Regulations and codes of practice
Familiarization with:
.1.1 the appropriate provisions of relevant international conventions;
.1.2 international and national codes;
.1.3 the IMO Manual on Oil Pollution; and
.1 .4 relevant tanker satety guid es .
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.2 Ship design and equipment of oil tankers
Familiarization with:
.2.1
.2.2
.2.3
.2.4
.2.5
.2 .6
2.7
piping, pumping. tank and deck arrangements:
types of cargo pumps and their application to various types of cargo;
tank clean ing, gas-freeing and inerting syst ems ;
cargo tank venting and accommodation ventilation:
gauging system s and alarms:
ca rgo heating systems; and
safety factors of electrical systems .
.3 Cargo characteristics

of cargo;
tank clean ing, gas-freeing and inerting syst ems ;
cargo tank venting and accommodation ventilation:
gauging system s and alarms:
ca rgo heating systems; and
safety factors of electrical systems .
.3 Cargo characteristics
Knowledge of chemical and physical properties of different oil cargoes .
. 4 Ship operations
Cargo calculations. Loading and dis charging plans. Load ing and dis c harge procedures including ship-ta-ship transfers. Check lists. Use of monit oring equipment. I mporta nce of proper sup@rvision of personn el. Gas-freeing operations a nd tank cleaning op@rations. Where appropriate. crude oil washing procedures and the operation and maintenance of inert gas systems. Contro l of entry into pump-rooms and enclosed spaces. Use of gas detecting and safety equipment. Load-on - top and proper ballasting and de-ballasting procedur@s. Air and water pollution prevention .
5 Repair and maintenance
Precautions to be taken before and during repair and maintenance wor k, including work affecting pumping, piping, electrical and control system s. Safety factors necessary in the performance of

5 Repair and maintenance
Precautions to be taken before and during repair and maintenance wor k, including work affecting pumping, piping, electrical and control system s. Safety factors necessary in the performance of hot work. Control of hot work and proper hot work procedures .
6 Rfterqency operations
The importance of developing ships' emergency plans. Cargo operations emergency shutdown. Action in the event of fa ilu re of services essential to cargo. Fire-fighting on oil tankers. Action following collision. stranding or spillage. First aid procedures and the use of resuscit at ion equipment. Use of breathing apparatus. Rescue from enclosed spaces.
Appendix 3 to regulation VII
Cbemical tapkers
The specialized training programme referred to in paragraph 2.2 of regulation VII appropriate to duties on chemical tankers shall provide theoretical and practical knowledge of th e following subjects:
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.1 RegulatioDs and codes of practice

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.1 RegulatioDs and codes of practice
Familiarization with relevant international conventions and with IHO. national and relevant international codes .

. 2 Ship design and equipment of chemical tankers
A brief description of specialized piping . pumping and tank arrangements, overflow control. Types of cargo pumps and their application to various types of cargo. Tank c le aning and gas-freeing systems. Cargo tan k venting and accommodation ventilation, airlocks. Gauging system s and alarms . Tank temperature control systems. The safet y fa cto rs of e l ectrical system s .

3 Cargo characteristics
Sufficient knowledge o f liquid chemical ca rgo c haracteristics t o enable the proper use of relevant in ternation a l codes .

4 Ship operatioDs
Cargo calculations. Loading an d di scharging p l ans. Loading and discharge procedure. Check lists. Use of moni tor ing equ i pment. Gas-freeing operations and tank c leaning operations (proper use of

4 Ship operatioDs

Cargo calculations. Loading an d di scharging p l ans. Loading and discharge procedure. Check lists. Use of moni tor ing equ i pment. Gas-freeing operations and tank c leaning operations (proper use of absorption and wetting agents and detergent s) . Us e and maint enan ce of inert atmospheres. Control of entry into pump - rooms and enclosed spaces. Use of detecting and safety equipment. Disposal of waste and washings

5 Repair and maintenance

Precautions to be taken before the r e pair and maintenance of pumping, piping, electrical and control syst ems .

. 6 z.ergency operations

The importance of developing ships' emerg ency plan s. Cargo operati ons emergency shutdown. Action i n the eve n t o f failure of services essential to cargo. fire -fi ghting on c hemi cal t ankers. Action following collision, stranding or spillage. fir st aid procedure and the use of r esus ci tati on and deco nt amination equipment. Use of breathing apparatu s. Res c ue f rom enclosed spaces.

Appendix 4 to regula tion VI I

Liquefied gas tankers

The spe cialized training programme referred to in par agraph 2.2 of

equipment. Use of breathing apparatu s. Res c ue f rom enclosed spaces.

Appendix 4 to regula tion VI I

Liquefied gas tankers

The spe cialized training programme referred to in par agraph 2.2 of regulation VII appropriate to the duties on liquefied gas tankers shall provide theoretical and practical knowledge of the following sub j ec ts:

.1 Regulations and codes of practice

familiariza t ion with relevant i nt e rnati ona l conventions and with IMO. national and relevant. i nter nati ona l codes .

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.2 Ship design and equipment of liquefied gas tankers, incuding familiarization with:

.2.1 Types of liquefied gas tankers;

.2.2 Cargo containment systems (construction, surveys);

.2.3 Cargo-handling equipment (pumps, piping syste ms) ;

.2.4 Cargo conditioning systems (warm-up, cool-down);

.2 . 5 Tank atmosphere control systems (inert g8S, nitrogen);

.2.6 Instrumen tation of cargo containment and handling systems;

.2.7 Fire-fighting systems; and

.2.8 Safety and rescue equipment .

.2 . 5 Tank atmosphere control systems (inert g8S, nitrogen);
.2.6 Instrumen tation of cargo containment and handling systems;
.2.7 Fire-fighting systems; and
.2.8 Safety and rescue equipment .
. 3 Fire - fighting
Advanced practical fire-fighting techniques and tactics applicable to gas tankers, including the use of water-spray systems .
4 Chemistry and physics
An introduction to basic chemistry and physics as it relates to the safe carriage of liquefied gases in bulk in ships:
.4.1 Properties and characteristics of liquefied gases and their vapours:
.4.1.1
.4.1.2
.4.1.3
.4.1.4
.4.1.5
.4.1.6
.4 .1 .7
.4.1.8
.4. 1.9
.4.1.10
.4.1.11
.4 .1.12
.4.1.13
.4.1.14
. 4.1.15
Definition of g8S;
Simple gas laws:
Gas equation;
Density of gases;
Diffusion and mixing in gases;
Compression of gases;
Liquefaction of gases;
Refrigeration of gases:
Critical temperature;
Practical significance of flashpoint;
Upper and lower explosive limits;
Auto-ignition temperature;
Compatibility of gases;
Reactivity; and
Polymerization. inhibitors .
.4.2 Properties of single liquidS
.4.2.1
.4.2.2
.4.2.3
.4.2.4
4.2.5
Densities of liquids and vapours;

Upper and lower explosive limits;
Auto-ignition temperature;
Compatibility of gases;
Reactivity; and
Polymerization. inhibitors .
.4.2 Properties of single liquidS
.4.2.1
.4.2.2
.4.2.3
.4.2.4
4.2.5
Densities of liquids and vapours;
Variation with temperature;
Vapour pressure and temperature;
Entalphy; and
Vaporization and boiling liq uids .
.4.3 Nature and properties of solutions
.4.3.1
.4.3.2
.4.3.3
Solubility of gases in liquids;
Miscibility between liquids and effects of
temperature change ;
Densities of solutions and dependen ce on temperature
and concentration;
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.4.3.5
.4.3.6
.4.3.7
.4.3.8
. 4.3.9
- 8 -
Effects of dissolved substances on melting and
boiling points:
Hydrates, formation and dispersion;
Hygroscopicit.y;
Drying of air and other gases;
Dew point; and
Low temperature effects .
.5 Health hazards
.5.1
.5.2
. 5 .3
Toxicity
.5.1.1
.5.1.2
.5.1.3
5.1.4
Hodes by which lique fi ed gases and their vapours may
be toxic;

Hygroscopicit.y;
Drying of air and other gases;
Dew point; and
Low temperature effects .
.5 Health hazards
.5.1
.5.2
. 5 .3
Toxicity
.5.1.1
.5.1.2
.5.1.3
5.1.4
Hodes by which lique fi ed gases and their vapours may be toxic;
Toxic properties of inhibitors and of pro ducts, of combustion of both materials of construction and of liquefied gases carried;
Acute and chronlc effe c ts of toxicity. systemic poisons and irritants;
Threshold limit value (TLV) .
Hazards of skin contact, inha lation and ingestion .
First aid and administering of antidotes .
. 6 Cargo contaiDment.
.6.1
.6.2
.6.3
.6.4
6.5
Principles of containment systems;
Rules;
Surveys;
Tank construction, materials, coatings, insulation; and
Compatibility .
.7 Pollution
.7.1
. 7.2
.7.3
. 7.4
Hazards to human life and to the marine environment;
Effect of specific gravity and solUbility;
Danger from vapour cloud drift; and
Jettisoning of cryogenic liquids .
.8 Cargo handling system
.8.1
.8.2
.8.3
.8.4
.8.5
.8.6
.8.7
.8.8
.8.9
.8.10
.8.11
.8.12
Description of main types of pumps and pumping arrangements and vapour return systems, piping systems and va l ves;

.8 Cargo handling system
.8.1
.8.2
.8.3
.8.4
.8.5
.8.6
.8.7
.8.8
.8.9
.8.10
.8.11
.8.12
Description of main types of pumps and pumping arrangements
and vapour return systems, piping systems and va l ves;
Ezplanation of pressure, vacuum, suction, flow, head;
Filters and strainers;
Ezpansion devices;
Flame screens;
Commonly used inert gases;
Storage, gene ration, distributi on systems;
Temperature and pressure monitoring syst ems;
Cargo vent sys tems;
Liquid re-cir culation and re-liquefaction systems;
Cargo gauging, instrumentation sys tem s and alarms ;
Gas detection and monitoring systems;
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. 8.13 C02 monitoring systems;
.8.14 Cargo boil-off systems: and
.8.15 Auxiliary systems
. 9 Ship operating procedures
.9.1
g. 2
.9.3
.9.4
g. 5
.9.6
.9.7
.9.8
.9.9
.9.10
Loading and disCharging preparat i ons and procedur es:
Check lists;
Cargo condition maintenance on passage and in harb our ;
Segregation of cargoe s and procedur es for cargo transfer;

g. 2
.9.3
.9.4
g. 5
.9.6
.9.7
.9.8
.9.9
.9.10
Loading and disCharging preparat i ons and procedur es:
Check lists;
Cargo condition maintenance on passage and in harb our ;
Segregation of cargoe s and procedur es for cargo transfer;
Changing carg oes . tank cleaning procedures;
Cargo sampling;
Ballasting and de-ballasting;
Warm up and cool down systems:
Warm up and gas-freeing procedures; and
Procedures for cool down of gaS-free system from ambient
temperature and safety pre ca utio ns i n vo lved .
. 1 0 Safety practices and equipment
.10.1 Function. calibrat i on and use of portabl e measuring
ins trwnent s;
. 10 .2 Fi r e-fightin g equipm ent and procedures;
.10.3 Breathing apparatus;
.10.4 Resuscitators;
.10. 5 Escape se t s;
. 10.6 Rescue equipment;
.10.7 Protective clothing and equi pmen t:
. 10.8 Entry into enclosed spac es;
.10.9 Precauti ons to be observed before and during repair and
maintenan c e of cargo and control systems;
.10.10 Supervision of personnel during potentially hazardou s
operations:
.10.11 Types and principle s o f ce rtified safe electr ical equipmen t ;
and
.10 . 12 Sources of igniti on .

maintenan c e of cargo and control systems;
.10.10 Supervision of personnel during potentially hazardou s
operations:
.10.11 Types and principle s o f ce rtified safe electr ical equipmen t ;
and
.10 . 12 Sources of igniti on .
. 11 Eaerqency procedures
.11.1
. 11.2
.ll .3
.ll.4
.11.5
The importance of developing ships' emergency plan s;
Emergency shutdown of cargo operations;
Emergen cy cargo valve closing systems:
Action in the event of failure o f systems or services
essential to cargo; and
Action following collision or stranding. spillage and
envelopmen t of ship in toxi c or flammable v a pour .
12 General principles of cargo operations
.12.1
.12.2
.12.3
.12 . 4
.12. 5
Inerting cargo tank and void spaces;
Tank cool down, loading;
Operations during loaded and ballasted voyages;
Discharging and tank stripping: and
Emergency pro c edures , including pre-planne d action in the
event of leaks. fire, collision, stranding , emergen c y carg o
discharge. personnel casualty."
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event of leaks. fire, collision, stranding , emergen c y carg o
discharge. personnel casualty."
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